Intergenerational Housing Misallocation and Fertility
Compact Theory and Quantitative Lifecycle Model

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Abstract

Children require goods and housing space. The space requirement matters because family-sized housing is harder to access than small rental housing: large units are concentrated in ownership, ownership requires down payments, and tax rules can induce older owners to retain houses whose marginal value to them is below the value to constrained young families. This draft has two parts. Part I gives a compact two-period model that isolates the mechanism and delivers goods-equivalent wedges for young access constraints, old-owner lock-in, and fertility. Part II embeds the same mechanism in a finite-life quantitative lifecycle model with income risk, tenure and house-size choice, down-payment constraints, transaction costs, bequests, child aging, and a single aggregate housing-services market.

The model combines a lifecycle fertility problem (Sommer, 2016; Doepke and Kindermann, 2019) with housing tenure choice, collateral constraints, and owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). The compact misallocation block follows the one-market floorspace logic in Coven et al. (2025): all housing services clear in one market, while tenure and tax rules determine who can occupy family-sized units.

Related Literature

This paper contributes to several strands of literature.

First, it contributes to the literature on fertility decline and its economic causes. A long tradition in economics treats fertility as a household choice in which children enter preferences and require resources (Becker, 1960; Becker and Lewis, 1973). Modern versions of this tradition emphasize lifecycle risk, the timing of births, and the household allocation of childrearing costs (Jones et al., 2010; Sommer, 2016; Doepke and Kindermann, 2019). Recent work on the U.S. fertility decline shows that falling birth rates are not well explained by a simple postponement margin or by one obvious period shock (Kearney et al., 2022), while demographic work interprets low fertility as part of a broader second demographic transition in family formation, marriage, and living arrangements (Myrskylä et al., 2013; Lesthaeghe, 2014). This paper follows the economics literature in treating fertility as a choice over the lifecycle, but it shifts attention to an input that is usually implicit: family-sized housing space. The relevant price of a child is not summarized by wages, childcare, or transfers alone when the child also requires housing services.

Second, it contributes to the empirical economics literature on housing markets, credit, and fertility. A central distinction in this literature is between current-period birth rates and completed fertility. Many influential papers identify effects on births, which are informative about mechanisms but need not by themselves show that final family size changes. Dettling and Kearney (2014) show that house-price increases reduce current births among nonowners and raise them among owners, consistent with offsetting housing-cost and home-equity effects. Lovenheim and Mumford (2013) find that housing wealth raises homeowners' fertility, and Cumming and Dettling (2024) show that mortgage-rate pass-through affects births through household cash flows. Hacamo (2021) finds that mortgage-market deregulation increased young households' home purchases and childbearing, with evidence pointing to access to space rather than income growth as the mechanism. These papers establish that housing and housing finance move fertility behavior, but most of the direct evidence is about births or birth hazards.

The evidence on completed fertility is smaller, but it is more direct than the current-birth literature alone. Simon and Tamura (2009) use U.S. Census data from 1940–2000 and find that higher rents per room are associated with lower fertility; their completed-fertility analysis for older households reinforces the main pattern. Clark (2012) provides an important caution: expensive U.S. housing markets delay first births by several years, but the paper does not find a clear decline in completed fertility. Japaridze and Sayour (2024) link unaffordable homeownership to delayed entry into motherhood and lower completed fertility. Two recent causal designs point in the same direction. Dettling and Kearney (2025) study the FHA and VA mortgage programs and find evidence that low-down-payment, long-term mortgage access increased births during the U.S. baby boom; their Census evidence on children ever born is consistent with higher completed fertility for groups with access to the programs. van Doornik et al. (2025) exploit randomized Brazilian housing-credit lotteries and find larger lifetime fertility effects when housing access arrives earlier in the childbearing years. This paper takes completed fertility as the model object and uses the birth-rate literature to discipline mechanisms rather than to stand in for final family size.

Third, it contributes to demographic and life-course research on housing and family formation. This literature documents that housing tenure, dwelling size, and residential transitions are jointly timed with childbearing. Mulder and Billari (2010) argue that homeownership regimes with difficult market entry are associated with low fertility. Evidence from Sweden and Finland links housing conditions and housing transitions to first births and higher-parity births (Ström, 2010; Kulu and Steele, 2013; Chudnovskaya, 2019). Tocchioni et al. (2021) show that the homeownership-parenthood association in Britain has weakened as housing access has become more difficult for young adults. This work supports the primitive that suitable housing is part of family formation, while also making clear that tenure is endogenous to fertility plans.

Fourth, the paper contributes to structural work on housing, tenure choice, and spatial allocation. The housing block builds on models in which tenure, collateral constraints, and owner-occupied housing shape household behavior (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). It also relates to spatial-equilibrium work in which housing supply and local prices allocate households across space (Rosen, 1979; Roback, 1982; Hsieh and Moretti, 2019). The closest recent structural paper is Couillard (2025), which studies joint location, house-size, and fertility choice and shows that the supply of large units matters for aggregate fertility. This paper instead focuses on the intergenerational allocation of family-sized space. Young families may value marginal space above its market user cost because they face rental-size limits, down-payment constraints, or thin access to ownership. Old incumbents may retain space whose private marginal value is below its value to constrained young families because tax rules subsidize staying put. The wedges below measure these allocation margins and connect them to completed fertility.

Contents

Part I

Compact Analytical Model

1 Environment

Agents. The economy is populated by overlapping generations of young and old households. A young household has type

$$i = (y_i, a_i),$$

where $y_i > 0$ is disposable income and $a_i \geq 0$ is entry wealth, inherited from the previous old generation. Conditional on entry, the young household chooses tenure, housing, completed fertility, and saving. Old households are indexed by j. An old incumbent is last period's young owner and enters old age with financial wealth, the house she bought, and an accrued taxable gain. She chooses consumption, how much housing to retain, and the estate she leaves at exit. An old renter carries no housing and no capital-gains wedge.

Preferences. Children raise both consumption needs and housing needs. In the compact model fertility is continuous. A young household choosing consumption c_i, housing h_i, and fertility n_i has utility

$$u_i^Y(c_i, h_i, n_i) = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i, \qquad \alpha, \vartheta, \chi, \kappa > 0. \tag{1}$$

The term χn_i is the goods requirement of children and κn_i is their housing-space requirement. A marginal child therefore uses the same housing stock as the parents.

Old households value consumption, housing, and bequests:

$$u^O(c_j^O, h_j^O, e_j) = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j), \qquad \gamma > 0, \tag{2}$$

where $e_j \geq 0$ is the estate and $\mathcal{B}$ is increasing and concave, possibly identically zero. Bequest motives and true old housing utility are preferences; they are not wedges.

Housing. There is a fixed stock $\bar{H}$ of divisible floorspace. Households occupy floorspace by renting or owning. The two tenure modes have different size limits:

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \qquad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\}, \qquad 0 < \bar{h}^R < \bar{h}^O. \tag{3}$$

This is the family-housing segmentation: small units are available to rent, but large family-capable units are mainly reached through ownership. All floorspace trades in one market. The user cost of housing is q and the asset price is P, linked by

$$q = \rho P, \qquad \rho = r + \delta + \tau^p, \tag{4}$$

where r is the interest rate, δ is depreciation, and τ^p is the property-tax rate. For a given user cost q, a higher property tax lowers the asset price P and therefore lowers the down payment needed to buy a given house.

Financing. A buyer can finance at most share $\phi \in (0, 1)$ of the purchase price. Buying h units therefore requires the cash down payment

$$(1 - \phi)Ph. \tag{5}$$

Renters cannot borrow. Households can save in a bond at return $1 + r$. The compact model treats the down-payment constraint as an access constraint: it limits the amount of owner housing a young household can occupy.

Capital-gains taxation and step-up basis. Selling a house triggers a capital-gains tax. Dying with the house gives a stepped-up basis, so the accrued gain is forgiven. In a stationary real model, the gain comes from nominal price growth. If dollar prices grow at rate ϖ, a house bought one period ago has a taxable paper gain even when its real value is unchanged. The real flow-equivalent tax wedge per unit sold is

$$\bar{\ell} = \tau^g \frac{P\varpi}{1+\varpi}, \tag{6}$$

after normalizing the period conversion factor to one. The wedge is not a preference and not a resource cost. It is a private tax liability avoided by holding the house until death.

2 Households

Conditional on entry, young household i chooses a tenure mode $m \in \{R, O\}$, housing, fertility, consumption, and savings. For $m \in \{R, O\}$,

$$\begin{aligned} W_i^m(q) = \max_{c_i, h_i, n_i, a_i'} \quad & \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \\ & + \beta \left[\mathbf{1}\{m = R\} V^R(a_{j'}) + \mathbf{1}\{m = O\} V^O(a_{j'}, h_i) \right] \end{aligned} \tag{7}$$

subject to

$$c_i + q h_i + a_i' = y_i + a_i, \tag{8}$$

$$h_i \in \mathcal{H}^m, \qquad a_i' \geq 0,$$

$$\mathbf{1}\{m = O\}(1 - \phi) P h_i \leq a_i, \tag{9}$$

$$c_i - \chi n_i > 0, \qquad h_i - \kappa n_i > 0.$$

Using $P = q/\rho$, the largest owner house feasible for type i is

$$H_i^O(q) = \min\left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi) q} \right\}, \qquad H_i^R = \bar{h}^R. \tag{10}$$

H_i^m is the effective family-housing cap. A renter is capped by the rental stock. A buyer is capped by the owner size limit or by collateral, whichever is shorter.

An old incumbent enters with financial wealth a_j, carried housing H_j^0, and an accrued gain. She chooses consumption, retained housing, and an estate:

$$\begin{aligned} V^O(a_j, H_j^0) = & \max_{c_j^O, h_j^O, e_j} \quad u^O(c_j^O, h_j^O, e_j) \\ & \text{s.t.} \quad c_j^O + e_j + q h_j^O = a_j + q H_j^0 - \bar{\ell}(H_j^0 - h_j^O), \qquad 0 \leq h_j^O \leq H_j^0. \end{aligned} \tag{11}$$

Selling one unit nets $q - \bar{\ell}$, so stepped-up basis acts as a subsidy to retaining the house. An old renter has no carried housing and no tax:

$$V^R(a_j) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \quad \text{s.t.} \quad c_j^O + e_j + q h_j^O = a_j, \qquad 0 < h_j^O \leq \bar{h}^R. \tag{12}$$

3 Entry and Equilibrium

A potential young household can enter the housing market or take an outside option with value $\bar{W}^E$. Entry has Type-I extreme-value shocks with scale κ_E. The inside value is

$$W_i^Y(q) = \max\{W_i^R(q), W_i^O(q)\},$$

and the entry probability is

$$\pi_i^E(q) = \frac{\exp\{W_i^Y(q)/\kappa_E\}}{\exp\{W_i^Y(q)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \tag{13}$$

Let F_O and F_R be the stationary measures of old incumbents and old renters, and let $\bar{M}$ be the mass of potential young households. Aggregate floorspace demand is

$$H^D(q) = \bar{M} \int \pi_i^E(q) h_i(q)\, \mathrm{d}G_Y(i) + \int h_j^O(q)\, \mathrm{d}F_O(j) + \int h_j^O(q)\, \mathrm{d}F_R(j). \tag{14}$$

The compact equilibrium clears the single stock:

$$H^D(q) = \bar{H}, \qquad P = q/\rho. \tag{15}$$

Stationarity requires current young owners to become next period's old incumbents, current young renters to become old renters, and births plus the renewal inflow to reproduce the next young cohort. The transition rules assign old estates to entry wealth and are otherwise left general; the local results below condition on the stationary cross-section.

4 Equilibrium Wedges

The model's sufficient statistics are marginal values of space in consumption goods. Let λ_i^m be the multiplier on the young household's budget, θ_i^m the multiplier on the size limit, and μ_i the multiplier on the owner's down-payment constraint. For renters,

$$\frac{\alpha(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R, \qquad \zeta_i^R = \frac{\theta_i^R}{\lambda_i^R} \geq 0. \tag{16}$$

A capped renter values one more unit of housing above the rent it would command.

For young owners, anticipation of the future capital-gains liability raises the effective price of owning family-sized space. Under interior saving and interior old retention,

$$q^O = q + \frac{\bar{\ell}}{1+r}. \tag{17}$$

The owner access gap is

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1-\phi)P}_{\zeta_i^{O,F},\ \text{down-payment component}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{owner-size component}} \geq 0, \tag{18}$$

so the owner's marginal value of space is

$$\frac{\alpha(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O. \tag{19}$$

When the owner size limit is slack, $\zeta_i^O = \zeta_i^{O,F}$ and the gap is a financing gap.

The old incumbent's interior retention condition is

$$\frac{\gamma c_j^O}{h_j^O} = q - \bar{\ell}. \tag{20}$$

She keeps marginal space she values below the market user cost because selling it realizes the taxable gain. An old renter has no such wedge and, at an interior choice, values space at q.

The fertility condition prices the marginal child in goods. In tenure mode m, with $q^R = q$ and q^O given by (17),

$$\underbrace{\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa(q^m + \zeta_i^m)}_{\text{full price of a child}}. \tag{21}$$

A child's space requirement is priced at the household's own shadow value of space. A binding access constraint therefore acts as an implicit child tax $\kappa\zeta_i^m$.

5 Efficiency and Fertility

The planner comparison holds the stationary cross-section, entry probabilities, tenure assignments, savings, and real size-limit constraints fixed at the instant of the variation. The planner counts true household preferences and true resource costs. It does not count the avoided capital-gains tax as a social benefit.

Proposition 1 (Local young-old housing misallocation)**.** *Consider a stationary equilibrium. Hold entry, tenure assignments, savings, and real size-limit constraints fixed. Take a young owner whose owner size limit is slack and whose financing constraint binds, so* $\zeta_i^{O,F} > 0$*, and an old incumbent at an interior retention choice. If one unit of existing housing can be reassigned from the old incumbent to the young owner at zero real implementation cost, the planner's goods-equivalent surplus is*

$$\underbrace{(q^O + \zeta_i^{O,F})}_{\textit{young buyer's marginal value}} - \underbrace{(q - \bar{\ell})}_{\textit{old incumbent's marginal value}} = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell}. \tag{22}$$

If this expression is positive, the competitive allocation does not solve the planner's conditional allocation problem.

The market price cancels. What remains are the young household's financing gap, the buyer's anticipated gains tax, and the old incumbent's realized gains tax. The two tax terms are transfers, not resource costs. If implementing the reassignment requires real cost r_i^F per unit, that cost subtracts from the surplus. If the down-payment constraint is a primitive technological enforcement constraint and no instrument can relax it, the variation is not feasible.

Old occupancy is not inefficient by itself. An old owner who keeps a large house because she values it, or because she values leaving bequests, is using housing for reasons the planner also counts. The inefficiency is the part of retention supported by the forgiveness of capital-gains taxes at death, paired with young households who value family-sized space above its market user cost.

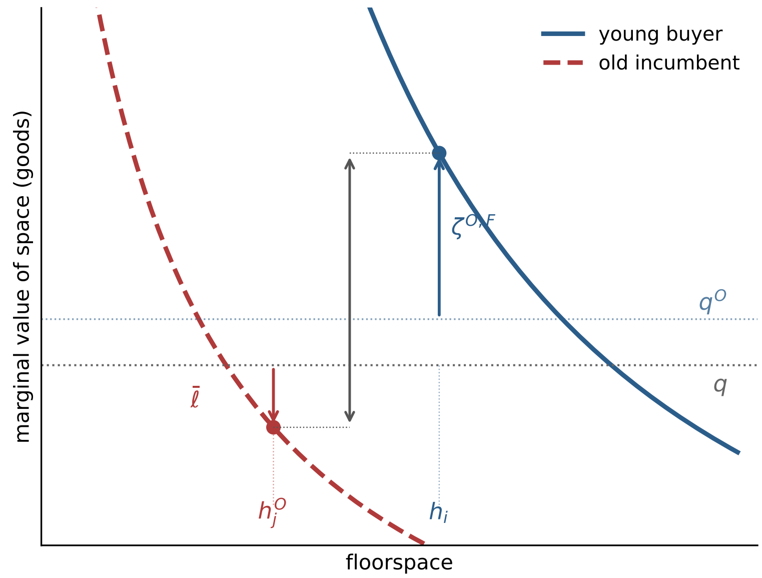

Figure 1: Local misallocation. A collateral-capped young buyer values marginal space at $q^O + \zeta^{O,F}$, while a locked-in old incumbent values retained space at $q - \bar{\ell}$. The vertical distance is the goods-equivalent surplus from a compensated marginal reassignment.

The same cap determines a fertility response. Fix a tenure mode and a household whose housing is pinned at the effective family-housing cap H. Concentrating the savings rule into the budget, fertility along the binding cap solves

$$\frac{\vartheta}{n} = \frac{(1+k)\chi}{w - q^m H - \chi n} + \frac{\alpha\kappa}{H - \kappa n},$$

where $w = y + a$ and k is the savings load on adult consumption. Differentiating with respect to the cap gives

$$\frac{\mathrm{d}n}{\mathrm{d}H} = \frac{\dfrac{\alpha\kappa}{(H-\kappa n)^2} - \dfrac{(1+k)\chi q^m}{(w - q^m H - \chi n)^2}}{\dfrac{\vartheta}{n^2} + \dfrac{(1+k)\chi^2}{(w - q^m H - \chi n)^2} + \dfrac{\alpha\kappa^2}{(H-\kappa n)^2}}. \tag{23}$$

Relaxing access raises fertility when the space-relief force dominates the budget-tightening force. A useful sufficient condition along every binding cap is

$$\chi \le \frac{(1+k)\kappa q^m}{\alpha}. \tag{24}$$

The derivative connects the misallocation wedge to fertility: policies that raise the effective family-housing cap matter only for households whose binding constraint they actually relax.

Part II

Quantitative Lifecycle Model

6 Purpose and Timing

The quantitative model embeds the same mechanism in a finite-life economy. It has the standard lifecycle structure: households age, face income risk, save, choose whether to rent or own, choose housing size, decide fertility, and eventually leave bequests. The housing market has one aggregate user cost and one asset price. Tenure matters because it changes the size menu, financing constraints, transaction costs, and tax exposure.

Time is discrete. Households enter adult life at age $a = 1$, retire after age J_R, and die at the end of age J. The individual state at the beginning of a period is

$$x = (b, z, d, a, n, s). \tag{25}$$

Here b is liquid wealth, $z \in \mathcal{Z}$ is idiosyncratic productivity, d is tenure and housing position, a is age, n is completed fertility, and s is the child-at-home state. Tenure belongs to

$$\mathcal{D} = \{R, O_1, \ldots, O_K\}. \tag{26}$$

R denotes renting. O_k denotes ownership of a house with physical size H_k, where $H_1 < \cdots < H_K$. Renters choose housing continuously up to $\bar{h}^R$; owners choose from the discrete ladder. The ladder is the quantitative version of the family-housing segmentation in Part I.

Income for working households is

$$y(a, z) = (1 - \tau_{\text{pay}}) w e_a z, \qquad z' \sim \Pi^z(\cdot|z), \qquad a \leq J_R, \tag{27}$$

and retired households receive pension income p^{ret}. The wage is common apart from age and productivity because the quantitative mechanism is housing access, tenure, and lifecycle wealth.

7 Preferences and Bequests

Let $m(n, s)$ be the number of dependent children living at home. Completed fertility n and current dependents differ because children eventually mature. Household needs are

$$\begin{aligned} \bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n m(n, s), \\ \bar{h}(n, s) &= \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{m(n, s) \geq 1\} + \bar{h}_n m(n, s). \end{aligned} \tag{28}$$

The first child can create a discrete housing-space jump; additional dependent children raise the required space linearly.

Flow utility is Stone-Geary over consumption and housing services, wrapped in CRRA:

$$u(c, \mathbf{h}; n, s) = \frac{\left[(c - \bar{c}(n, s))^{\alpha} (\mathbf{h} - \bar{h}(n, s))^{1-\alpha}\right]^{1-\sigma}}{1 - \sigma} + \psi_{\text{child}} m(n, s), \tag{29}$$

with utility equal to $-\infty$ when either residual is nonpositive. Housing services differ by tenure:

$$\mathbf{h} = \begin{cases} h^R, & d = R, \\ \chi_O H_k, & d = O_k, \end{cases} \qquad \chi_O > 0. \tag{30}$$

In market clearing, owners demand the physical quantity H_k, not the service quantity $\chi_O H_k$.

At death, households receive warm-glow bequest utility. Let W^B be bequeathable wealth after liquidation costs and taxes. Bequest utility is

$$\mathcal{B}(W^B, n) = \theta_0(1 + \theta_n n)\frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1-\sigma}. \tag{31}$$

The parameter θ_n lets completed fertility affect the desired estate even after children have left the household.

8 Housing, Finance, and Taxes

There is one aggregate housing-services market. The user cost q and asset price P satisfy

$$q = (r + \delta + \tau^p)P. \tag{32}$$

Housing supply is upward sloping:

$$H^S(q) = H_0 \left(\frac{q}{\bar{q}}\right)^\eta. \tag{33}$$

The fixed-stock compact model is the limiting case $\eta = 0$ around the benchmark stock.

Renters choose c, h^R, b' subject to

$$c + qh^R + b' = Rb + y(a, z), \qquad 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0, \tag{34}$$

where $R = 1 + r$. Owners of rung H_k choose c, b' subject to the owner branch budget and the borrowing limit described below. Owners pay maintenance and property taxes through $(\delta + \tau^p)PH_k$.

At origination, a buyer can finance at most share ϕ of house value, so the required down payment is

$$(1 - \phi)PH_k. \tag{35}$$

The post-purchase liquid position must satisfy

$$b' \ge -\phi PH_k. \tag{36}$$

An incumbent owner who keeps the same house is not forced to delever after a price change:

$$b' \ge \min\{b, -\phi PH_k\}. \tag{37}$$

This convention constrains origination but lets incumbency carry state dependence.

Housing adjustment is costly. Let ψ^s be a proportional sale cost. When the capital-gains block is active, the state of an owner also includes a tax basis B^p. A sale of house H_k triggers tax

$$\mathcal{T}^g(PH_k, B^p) = \tau^g \max\{PH_k - B^p - \bar{g}, 0\}, \tag{38}$$

where $\bar{g}$ is the statutory exclusion. Net sale proceeds are

$$\Omega(H_k, B^p) = (1 - \psi^s)PH_k - \mathcal{T}^g(PH_k, B^p). \tag{39}$$

At death, stepped-up basis resets the heir's basis to market value, so accrued gains are not taxed. This is the quantitative counterpart of the old-owner lock-in wedge $\bar{\ell}$ in Part I.

9 Fertility and Child Aging

Fertility is chosen during the fertile window

$$a \in \{A_f^{\text{start}}, \dots, A_f^{\text{end}}\}.$$

A childless household chooses

$$n' \in \mathcal{N} = \{0, 1, \dots, \bar{n}\}. \tag{40}$$

The option $n' = 0$ means no birth in the current period. A positive choice fixes completed fertility forever and moves the household into the first dependent-child state. A household that reaches the end of the fertile window without a positive choice remains childless.

Let $s = 0$ denote no dependent children and let $s = 1, \dots, S_c$ denote dependent-child stages. After a positive fertility choice, the household enters $s = 1$. Children age stochastically through the stages according to Π^c. While children are at home, $m(n, s) = n$; after maturity, $m(n, s) = 0$. Completed fertility still matters through bequest utility.

10 Household Problem

Within the period, the household first faces fertility if eligible, then chooses tenure and housing position, then chooses consumption, rental housing, and saving. Let the continuation value after income risk and child aging be

$$\bar{V}_{a+1}(b', d', z, n, s) = \sum_{z'} \sum_{s'} \Pi^z(z'|z) \Pi^c(s'|s, n) V_{a+1}(b', z', d', n, s'). \tag{41}$$

The terminal value is

$$V_{J+1}(b, z, d, n, s) = \mathcal{B}(W^B, n).$$

Conditional on renting, the branch value is

$$\begin{aligned} V_a^R(b, z, n, s) = \max_{c, h^R, b'} \quad & u(c, h^R; n, s) + \beta \bar{V}_{a+1}(b', R, z, n, s) \\ \text{s.t.} \quad & c + q h^R + b' = Rb + y(a, z), \\ & 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0. \end{aligned} \tag{42}$$

Conditional on owning rung H_k after any adjustment, the branch value is

$$\begin{aligned} V_a^{O_k}(\tilde{b}, z, n, s) = \max_{c, b'} \quad & u(c, \chi_O H_k; n, s) + \beta \bar{V}_{a+1}(b', O_k, z, n, s) \\ \text{s.t.} \quad & c + (\delta + \tau^p) P H_k + b' = R\tilde{b} + y(a, z), \\ & b' \ge \underline{b}_k^O. \end{aligned} \tag{43}$$

The lower bound $\underline{b}_k^O$ is the origination or incumbent constraint in (36)–(37), depending on whether the household buys or keeps the current house.

Let $H(R) = 0$ and $H(O_k) = H_k$. The liquid wealth available after housing adjustment is

$$\tilde{b}(b, d, d') = b + \mathbf{1}\{d \in \mathcal{O},\ d' \ne d\} \Omega(H(d), B^p) - \mathbf{1}\{d' \in \mathcal{O},\ d' \ne d\} P H(d'). \tag{44}$$

Without the capital-gains block, $\Omega(H(d), B^p) = (1 - \psi^s)PH(d)$.

The tenure branch value for feasible d' is

$$\widetilde{V}_a^T(d'|b,z,d,n,s) = \begin{cases} V_a^R(\tilde{b}(b,d,R),z,n,s), & d' = R, \\ V_a^{O_k}(\tilde{b}(b,d,O_k),z,n,s), & d' = O_k. \end{cases} \tag{45}$$

Tenure and owner-rung choices have Type-I extreme-value shocks with scale κ_T:

$$\pi_a^T(d'|x) = \frac{\exp\{\widetilde{V}_a^T(d'|x)/\kappa_T\}}{\sum_{\hat{d}\in\mathcal{D}(x)} \exp\{\widetilde{V}_a^T(\hat{d}|x)/\kappa_T\}}, \tag{46}$$

and the inclusive tenure value is

$$V_a^T(x) = \kappa_T \log \sum_{\hat{d}\in\mathcal{D}(x)} \exp\{\widetilde{V}_a^T(\hat{d}|x)/\kappa_T\}. \tag{47}$$

For a childless household in the fertile window, the value of fertility option n' is

$$\widetilde{V}_a^F(n'|b,z,d) = V_a^T(b,z,d,n',s(n')), \qquad s(n') = \mathbf{1}\{n' > 0\}. \tag{48}$$

With Type-I extreme-value shocks of scale κ_F,

$$\pi_a^F(n'|b,z,d) = \frac{\exp\{\widetilde{V}_a^F(n'|b,z,d)/\kappa_F\}}{\sum_{\hat{n}\in\mathcal{N}} \exp\{\widetilde{V}_a^F(\hat{n}|b,z,d)/\kappa_F\}}. \tag{49}$$

Outside the fertile window, or after a positive fertility choice, this stage is degenerate and $V_a = V_a^T$.

11 Stationary Equilibrium

Let g be the stationary distribution of adult households over states. Adult mass is normalized to one for the benchmark quantitative exercise:

$$\int dg(x) = 1. \tag{50}$$

This normalization fixes the demographic envelope and lets fertility change household composition, tenure, and housing demand.

Definition 1 (Stationary equilibrium)**.** *A stationary equilibrium is a user cost and asset price (q, P), value functions and policies, tenure and fertility probabilities, a stationary distribution g, and aggregate housing quantities (H^D, H^S) such that:*

1. *households solve the branch problems (42)–(43), tenure choice is given by (46), fertility choice is given by (49), and terminal utility is (31);*
2. *the distribution g is invariant under policies, income transitions, fertility shocks, child aging, deterministic aging, death, and entry of new adult households;*
3. *aggregate physical housing demand is*

$$H^D = \int \left[\mathbf{1}\{d = R\}h^R(x) + \sum_{k=1}^{K} \mathbf{1}\{d = O_k\}H_k \right] \mathrm{d}g(x); \tag{51}$$

4. *the housing market clears and the user-cost relation holds:*

$$H^D = H^S(q), \qquad q = (r + \delta + \tau^p)P; \tag{52}$$

5. *payroll taxes, property taxes, capital-gains taxes, pensions, and any rebates satisfy the government accounting rule specified for the calibration or counterfactual.*

12 Calibration Objects

The quantitative model should be disciplined by the margins that identify the housing-fertility mechanism. The table gives the paper-facing organization of the calibration; it is not a results table.

Table 1: Parameter blocks and target moments

Block	Parameters	Main moments
Preferences and fertility	$\beta, \sigma, \alpha, \psi_{\text{child}}, \kappa_F$	completed fertility, childlessness, timing of first birth, wealth profiles
Child costs	$\bar{c}_n, \bar{h}_{\text{jump}}, \bar{h}_n$	rooms or housing services around first birth and additional births
Tenure and housing services	$\chi_O, \bar{h}^R, \{H_k\}, \kappa_T$	ownership by age, parent ownership gap, renter-owner size gap
Credit and adjustment	ϕ, ψ^s	ownership transition, liquid-wealth constraints, sale and downsizing rates
Bequests	$\theta_0, \theta_n, \theta_1$	old-age ownership, wealth at death, parent-childless saving differences
Capital gains and lock-in	$\tau^g, \bar{g}, B^p$	old-owner retention, downsizing around gains exposure, tax basis moments
Housing supply	$H_0, \eta, \bar{q}$	housing expenditure share, aggregate housing services, price response

The moment table should separate model moments from calibration targets. The natural blocks are fertility, lifecycle ownership, housing-size responses to children, old-owner retention, wealth and bequests, and market-clearing quantities.

13 Counterfactuals

The policy experiments should map directly into the primitive access margins:

1. relax down-payment constraints by increasing ϕ or giving targeted down-payment resources to young families;
2. expand family-sized rental access by increasing $\bar{h}^R$ or reducing the premium on large rental units;
3. reduce transaction costs ψ^s;
4. reform capital-gains taxation and step-up basis by changing τ^g, $\bar{g}$, or the treatment of gains at death;
5. increase housing supply through H_0 or η;

6. combine supply expansion with access policies to separate price effects from allocation effects.

Each counterfactual should report fertility, childlessness, ownership, young-family housing, old-owner retention, prices, and welfare. The compact model says where the effects should be strongest: households with positive family-housing access gaps and policies that pass through to the binding cap.

14 What Carries Over from the Compact Model

The quantitative model changes the state space but not the mechanism. In the compact model, a constrained young household prices a child's housing need at $q^m + \zeta_i^m$, while an old locked-in owner values retained marginal space at $q - \bar{\ell}$. In the quantitative model these are not single closed-form objects for every state, but they remain the diagnostic objects:

$$\text{young family-space shadow value} \quad \text{and} \quad \text{old retention shadow value.}$$

The calibration must therefore discipline not only aggregate fertility and aggregate ownership, but also the distribution of housing space by age, tenure, and child status. A model that matches total ownership while putting too much family-sized housing with unconstrained households would miss the mechanism.

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